

Name: Bhumiika Patwal ; Section- 'A' ; GEU (Deemed)
Taught By: Mr. Hemant Singh Pokhariya

Unit-1 (LD+CO)

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Implement :

1. Ex-OR gate using minimum no. of NAND gates.
2. Ex-NOR gate using minimum no. of NOR gates.
3. $Y = A(B+C)$ using minimum no. of NAND gates.

1. Ex-OR. Output = 1 when odd no. of 1's in input.

SOP Sum of Product

- Minterm: It is the product term which contains all the variables either in complemented form ($\bar{x}$) or uncomplemented form (x). These two are called literal.

$$\text{SOP } Y = A + \bar{A}B$$

$$= A \cdot 1 + \bar{A}B = A(B + \bar{B}) + \bar{A}B = AB + A\bar{B} + \bar{A}B$$

Standard / Canonical SOP $Y = AB + A\bar{B} + \bar{A}B$.

It is sum of minterms.

Standard / Canonical POS: It is product of maxterms.

- Maxterm: It is sum term which contains all the variables either in complemented or uncomplemented form.

$$\text{POS} = \Pi (\text{Maxterms})$$

$$\text{O/P} = 0 ; X = 0 \quad \bar{X} = 1$$

$$\text{SOP} = \Sigma (\text{Minterms})$$

$$\text{O/P} = 1 ; X = 1 \quad \bar{X} = 0$$

Total combinations = 2^n , $n \rightarrow$ no. of variables

Ex-OR. gate

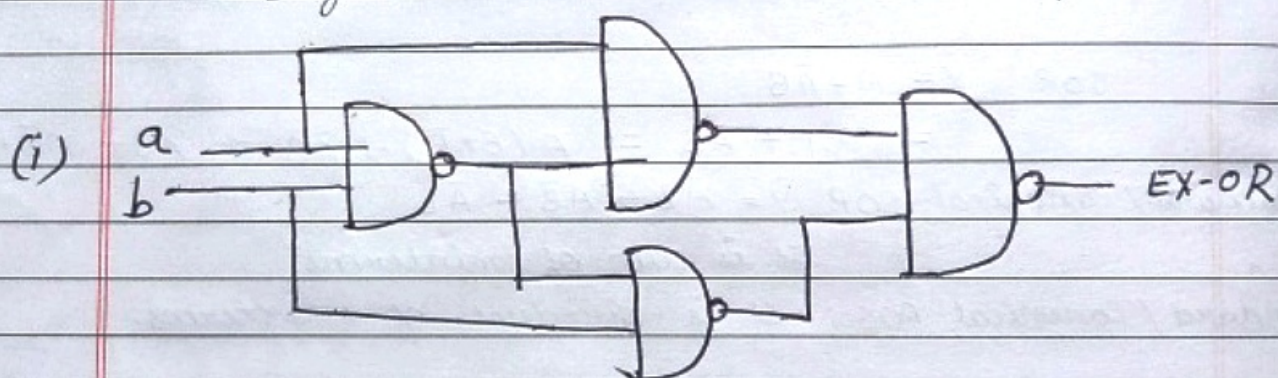
	A	B	Y
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

minterm $\rightarrow$ maxterm

$$Y = \sum m(1,2) = \bar{A}B + A\bar{B}$$

$$Y = \prod M(0,3) = (A+B) \cdot (\bar{A} + \bar{B})$$

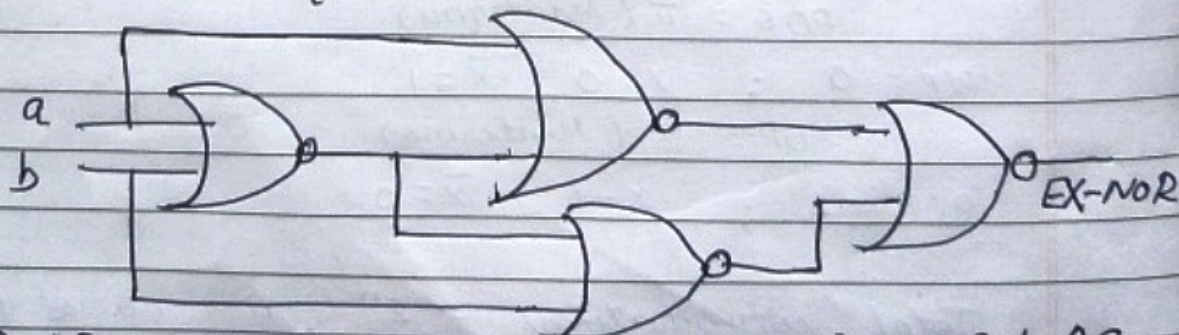
~~OR gate~~



by Duality Theorem

(ii) ~~AB~~ EX-OR $\xrightarrow{\text{Duality}}$ EX-NOR
NAND $\xrightarrow{\text{Duality}}$ NOR

Replace ~~NAND~~ NAND by NOR



(iii) AND-OR $\xrightarrow{\text{Duality}}$ NAND-NAND so $Y = A \cdot (B+C) = AB + AC$

